

Bottled Kraken



User Guide for Version 3.0

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1. Programme purpose and basic idea

Bottled Kraken is a desktop application for processing images and PDFs with a focus on OCR for difficult, historical, form-like, or handwritten influenced. The application combines classic OCR with manual correction, image preparation and optional AI-supported post-processing steps.

The program is not a one-click tool for "perfect text", but a working environment for controlled, step-by-step improvement. This is particularly useful for archive material, civil status documents, administrative documents, forms, address lists, hard-to-read scans and heterogeneous digitized copies.

Guiding principle: First the image and line structure is mastered, then the text is recognized, then improvements are made only where it makes sense from a technical and organizational point of view.

2. Interface at a glance

The interface is divided into several workspaces: queue, image preview, recognized lines/words, menus/toolbars, status display and additional dialogs for help, image editing, whisper, AI tools and export.

Area	Function	Practical benefits
Queue	Manages loaded images, PDFs, and project files.	Multiple files can be collected and prepared and processed one after the other.
Quick view	Shows current image or current PDF page.	Basis for visual inspection, overlay boxes and manual decisions.
Line/Word View	Lists recognized lines and words.	Here, OCR results are checked, moved, supplemented or deleted.
Menus/Buttons	Start OCR, model selection, export, help, AI and language functions.	Quick access to central work steps.
Status/Log	Shows ongoing operations, errors, and progress.	Useful for traceability and error analysis.

3. Supported inputs and files

Typical image formats as well as PDF files are supported. In addition, there are project files that can be used to save a processing status and load it later.

- Bildformate: PNG, JPG, JPEG, BMP, TIF, TIFF, WEBP
- PDF files as multi-page input sources
- Project files in JSON format
- Text and JSON files for line imports

When a PDF is loaded, pages are prepared or rendered internally so that they can be processed like individual image pages. This is especially important for files, scan collections and form bundles.

4. The standard workflow step by step

1. Load file or PDF.
2. Use image editing if necessary to improve cropping, contrast, or separation.
3. Select Segmentation Model and Recognition Model.
4. Start Kraken OCR.
5. Check, move, add, delete or correct recognized rows using overlay boxes.
6. Optionally, use multi-OCR when comparing multiple recognition models.
7. Optionally, use AI post-processing – locally or via OpenRouter, depending on the purpose.
8. Optionally, re-dictate individual lines with faster-whisper or import lines from files.
9. Save the project or export results.

Important: The standard path is not always linear. In practice, you often jump back and forth between preview, line structure, image editing and export checking.

5. Image editing before the OCR run

Image processing is not only used for optics, but also for structuring the material. Even small improvements to cropping, separation or contrast can significantly stabilize the OCR result.

- Pruning Areas of Interest
- Separation of double pages, form halves or disturbing edge areas
- Grayscale and contrast adjustment
- Preparation of poorly scanned or unevenly exposed originals
- Apply to current image, flagged images, or all images

Edited images are created as new output files and inserted back into the workflow. As a result, the original input remains untouched and the editing step is traceable.

6. Models: Segmentation and Recognition

Bottled Kraken deliberately separates between layout/line recognition and actual text recognition. For good results, both types of models must match the material.

Model Type	Task	Typical mistake in the wrong choice
Segmentation	Recognizes rows, baselines, and reading order.	Lines are contracted, fragmented, or read in the wrong order.
Recognition	Reads characters and words from recognized lines.	Letters, names, numbers or abbreviations are systematically recognized incorrectly.

Note: For automatic detection of the Kraken models, they must be located in the same directory (folder) as the Kraken program.

7. Kraken OCR in Practice

Kraken is the OCR/ATR basis of the application. It is particularly suitable for historical prints, more difficult layouts and specialized models. In Bottled Kraken, the Python library is addressed directly; however, the classic work steps are translated into the interface.

- Segment page layout and rows
- Reading Rows with Recognition Model
- Displaying Results in Line/Word Structure
- Further processing in export, correction or AI steps

Practical rule: If OCR is unexpectedly bad, first check segmentation, cropping, reading direction and model fit – only then think about "AI rescue".

8. Lines, words, and overlay boxes

Rows and overlay boxes are the central mechanism for technical control. Here, targeted corrections can be made if the automatic layout analysis was not accurate enough.

- Move a row up or down
- Delete row
- Create a new row above or below an existing row
- Move row to target position or swap with another row
- Draw, edit, or delete an overlay box

These features are especially useful for forms, column layouts, hard-to-read handwriting, and OCR results with incorrect line demarcation.

9. Multi-OCR

Multi-OCR unleashes multiple recognition models on the same file or selection, creating multiple OCR variants. These variants can be compared or merged.

Multi-OCR makes sense when different models have different strengths – for example, one model for historical print types and another for columns of numbers or form text.

Step	Description
1	Select recognition models (multiple selection possible).
2	Optionally, keep the segmentation/baseline model.
3	Create multiple OCR runs.
4	Merge results locally or share them with remote AI tools.

Note: Restriction in on-premises AI operations: On-premises AI usage is not intended for multi-OCR follow-up and not for the full pipeline in version 3.0. Multi-OCR follow-up with AI and the combined full pipeline are part of the OpenRouter/Remote AI path.

10. Local AI and LM Servers

For local AI post-processing, Bottled Kraken typically expects an OpenAI-compatible local model server. Typical examples are LM Studio, Ollama, GPT4All, Jan, text-generation-webui or LocalAI. In practice, a base URL is usually entered with /v1.

- LM Studio: often the most convenient GUI entry point
- Ollama: lean local service with OpenAI compatibility under /v1
- GPT4All: Desktop solution with local API server
- Jan, text-generation-webui and LocalAI: more local or self-hosted variants

In local AI operation, the focus is on local post-processing steps and the local generation of PostgreSQL JSON or Neo4j JSON.

Note: In local AI model operation, multi-OCR AI follow-up and the entire full pipeline are not the intended way in this version. This is what the OpenRouter/Remote AI rail is for.

The version based on version 3.0 adds an LM OCR mode to local AI operations as well as conservative internal token boundaries for LM revision.

- LM revision means that
 - (a) only the currently selected row,
 - (b) the marked lines, or
 - (c) all recognized rows are revised.
- In the LM revision, the locally loaded AI model uses the already generated Kraken OCR overlay boxes and independently transcribes the text inside, compares them with the already recognized lines/words and corrects errors.
- There are three gradations:
 - (a) Print: the LM works only inside the overlay box,
 - (b) Handwriting: the LM works beyond the boundaries of the overlay box (to be able to recognize more text),
 - (c) Mixed text: the LM works with less tolerance outside the overlay box than is the case with "handwriting".
- LM OCR performs a complete page scan with a local model and assigns the results exclusively to the already existing overlay box lines.
- LM OCR requires existing overlay boxes; already recognized Kraken lines and words do not serve as the primary basis.
- While LM OCR is running, the application shows a hint dialog with loading animation instead of a classic progress bar.
- The normal LM revision limits the response length internally depending on the type of request: single line or decision request up to 220 tokens, block request up to a maximum of 900 tokens, page-related assignment up to a maximum of 1200 tokens.

Quantization Code	Bits Per Weight (effective)	Relative size	Quality Description	Recommended use case
Q8_0	8.0	Largest	Almost lossless	Systems with >12 GB VRAM; maximum quality is more important than speed.
Q6_K	~6.6	Very large	Extremely low quality loss	Excellent choice if it fits completely into the VRAM buffer.
Q5_K_M	~5.5	Large	Very low quality loss	Recommended sweet spot for quality on 8GB systems.
Q4_K_M	~4.5	Medium	Balanced quality	Recommended sweet spot for performance on 8 GB systems.
Q3_K_M	~3.4	Very small	Noticeable loss of quality	Only in case of extreme resource constraints or for experiments.
Q2_K	~2.6	Smallest	Severe loss of quality	Not recommended for serious use.

Quantization Table (<https://www.martinkaessler.com/lokale-ki-modelle-auf-pcs-mit-8-gb-vram/>)

11. OpenRouter and Remote AI Tools

OpenRouter is used in Bottled Kraken as an external, OpenAI-compatible remote AI access. It can be used for OCR merge, PostgreSQL JSON, Neo4j JSON and the full pipeline.

Field	Significance	Example
Provider	Type of service	openrouter
API key	Access key	personal OpenRouter key
Base URL	Basis-Endpoint	https://openrouter.ai/api/v1
Model	Language model used	an activated OpenRouter model
Temperature	Controls Variance of Output	0.2 for more stable JSON outputs
Timeout	Maximum waiting time per request	90 seconds

OpenRouter is practical when strong cloud models are needed or when several models or providers are to be tried out quickly. In return, the data leaves the purely local area.

12. OCR-Merge, PostgreSQL-JSON, Neo4j-JSON und Full Pipeline

The AI tools are geared towards four typical aiming modes. Which mode makes sense depends on whether a better target text, relational data sets, graph structures or a combined overall process are needed.

Mode	Result	When it makes sense
OCR Merge	A sanitized target text from multiple OCR variants.	If several OCR variants are to be merged into a plausible reading text.
PostgreSQL-JSON	A JSON with a document- and table-oriented structure.	When data is to be transferred to relational tables or ETL processes later.
Neo4j-JSON	A JSON with entities and relationships.	When people, places, organizations, and relationships are to be modeled as a graph.
Full Pipeline	Merge plus structured AI results in one flow.	If the remote AI path is to execute several steps in a row as compactly as possible.

Important: The Full Pipeline is assigned to Remote AI/OpenRouter operation. In local AI operation, it is not intended as a complete path in version 3.0.

13. PostgreSQL JSON in Detail

The application creates a relationally oriented JSON object. The goal is a comprehensible intermediate structure that can be further processed in databases, scripts or QA processes.

Typical top-level areas include document metadata, people, places, streets, years, organizations, and references. Unsafe values should be treated as unsafe and not blindly adopted automatically.

- Suitable for data cleansing and pre-structuring
- Helpful for later tabular models or review workflows
- Particularly useful for forms, lists and administrative files

14. Neo4j JSON in Detail

Neo4j JSON is suitable when relationships are more important than pure table logic. Typical elements are nodes (e.g., people, places, documents, organizations) and edges (e.g., residing in, related to, mentioned, active at).

This format is particularly interesting for network analysis, relationship research, origin and person analyses.

15. Voice input with faster-whisper

Faster-whisper is used locally to re-record individual lines via the microphone. This is not a full replacement for OCR, but a precision tool for difficult passages.

1. Deploy or download the Whisper model.
2. Select the microphone.
3. Select a line.

4. Start recording and speak.
5. Review and apply transcript.

Note: Name fields, number sequences, individual damaged lines or short form information can often be dictated faster and more securely than retyping.

16. Row import from TXT and JSON

Recognized or externally prepared lines can be imported from TXT or JSON files – for the current image, selected images, or all images. The file names must be assigned appropriately via the base name.

This is practical when external correction processes, transcription aids or already structured preparatory work are to be integrated.

17. Save and reload projects

With project files, a work state can be saved and reopened later. This is particularly important for longer machining series, team coordination or audit-proof intermediate steps.

- Save project
- Save project as
- Project Barn
- Check missing files on load

18. Export formats

Bottled Kraken supports multiple export channels – depending on the technical goal. In addition to classic text formats, there are structure-oriented OCR outputs and PDFs.

Format	Benefits
TXT	Simple reading text or further processing in editor/script.
CSV	Tabular evaluation or import into data tools.
JSON	Structured further processing and exchange.
ALTO	OCR/layout exchange format.
hOCR	HTML-like OCR representation with position information.
PDF	Document-level output format.

In addition, AI merge texts, PostgreSQL JSON and Neo4j JSON can be exported separately.

19. Typical error situations and their significance

Version 3.0 contains numerous translations for error messages in German, English and French. Nevertheless, the classification of the content is important so that the reaction is not only linguistically but also technically correct.

- No recognition or segmentation model loaded: OCR cannot start meaningfully.
- LM server unreachable: Check URL, running model, and API compatibility.
- OpenRouter authentication failed: Check API key and model entry.

- Invalid JSON from the AI: Check model selection, temperature, and prompt stability.
- No OCR texts available: Check selection, status or pre-processing.
- No recognition models found for Multi-OCR: Check model folder or scan.

Recommendation for practice: Always check input data, model selection, status display and server path first. Many supposed "AI errors" are in fact selection or configuration errors.

20. Important keyboard shortcuts

Abbreviations	Function
Ctrl+S	Save project
Strg+Shift+S	Save project as
Ctrl+I	Project Barn
Ctrl+E	Export
Ctrl+Q	Quit Program
Ctrl+K	Start Kraken OCR
Ctrl+P	Stop OCR
Ctrl+L	Start LM revision
Ctrl+M	Dictate lines via microphone
F1–F7	Quick accesses for Help, Models, Server, Whisper and Log

21. Typical error situations and their significance

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- LM server unreachable: Check URL, running model, and API compatibility.
- OpenRouter authentication failed: Check API key and model entry.
- Invalid JSON from the AI: Check model selection, temperature, and prompt stability.
- No OCR texts available: Check selection, status or pre-processing.
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Recommendation for practice: Always check input data, model selection, status display and server path first. Many supposed "AI errors" are in fact selection or configuration errors.

22. Data protection

Data protection here depends primarily on whether a step really remains local or whether data is transmitted to external services.

Operating mode	Data protection effect
Local	Texts typically remain on the device or their own infrastructure – provided there is no external forwarding, telemetry or cloud route.

OpenRouter/ Remote	Texts leave the local area; in addition, other providers may be involved depending on the model selected.
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For personal, confidential or archivally sensitive content, the local route should be the standard, provided that the technical task allows it. In the case of OpenRouter or other external services, the legal basis, responsibilities, possible order processing and international data flows must be clarified in advance.

23. Legal

Legally, at least three levels must be distinguished: the license of the application, the license of the integrated components or model servers and the rights to the processed content itself.

- The Bottled Kraken application base is to be considered separately from OCR, Whisper and AI models under licensing law.
- Model licenses can be permissive, copyleft-based, or usage-restricted.
- Even in the case of permitted software use, copyright, archive, data protection or service law limits may apply to the processed sources.

25. Quality control in everyday life

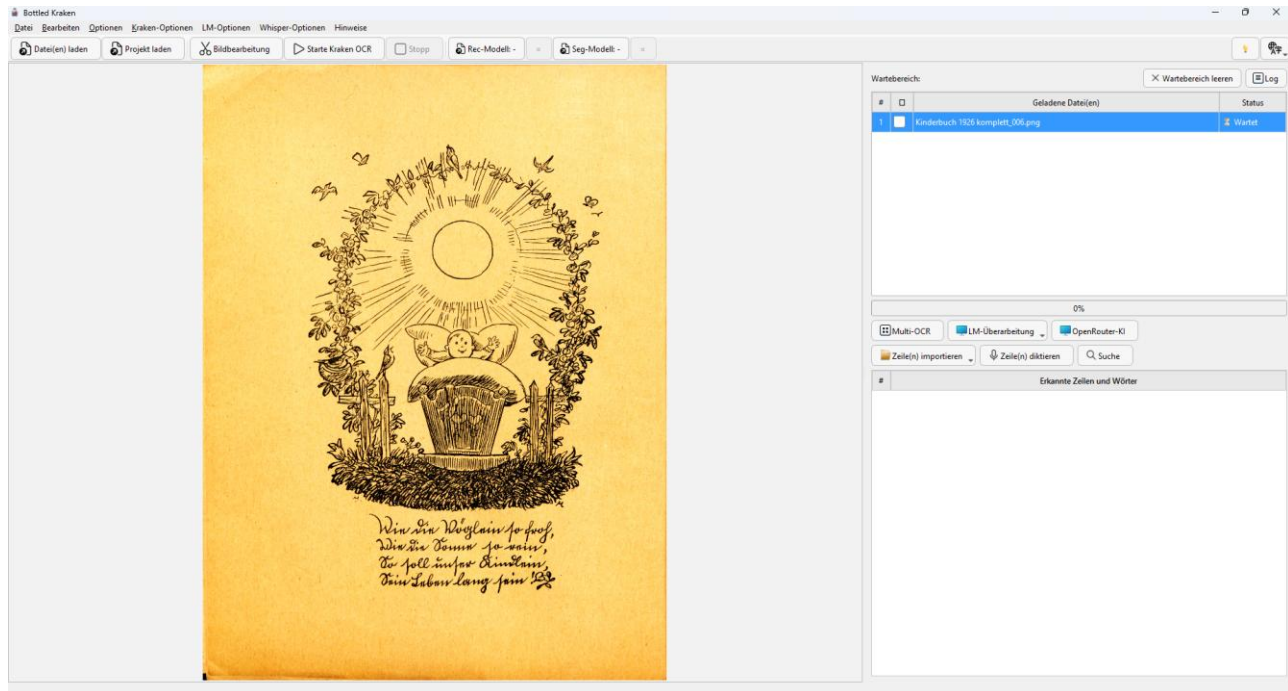
Good results do not come from a maximum of functions, but from controlled decisions in the right places.

1. Check input quality.
2. Take segmentation seriously before recognition.
3. Manually check the plausibility of the line structure.
4. Never treat AI results as truth without verification.
5. Always check JSON outputs on a random basis.
6. Version project statuses cleanly for longer work.

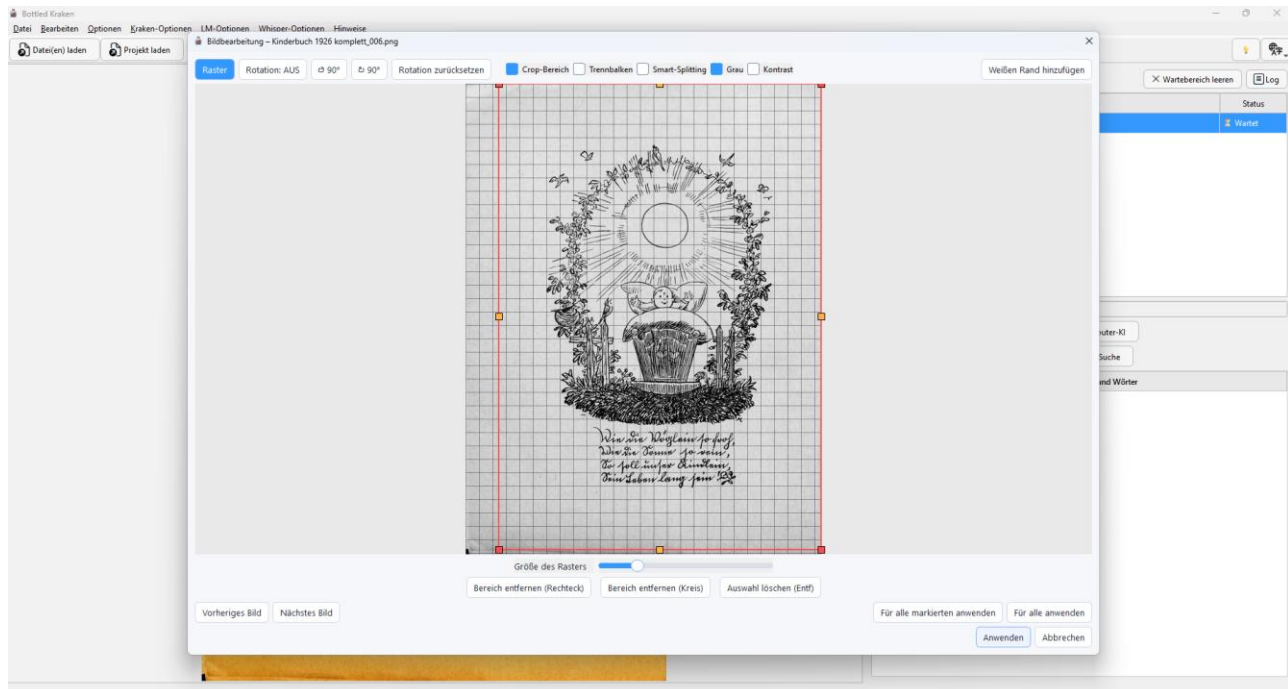
26. Practical example

The following is a short working process in picture form.

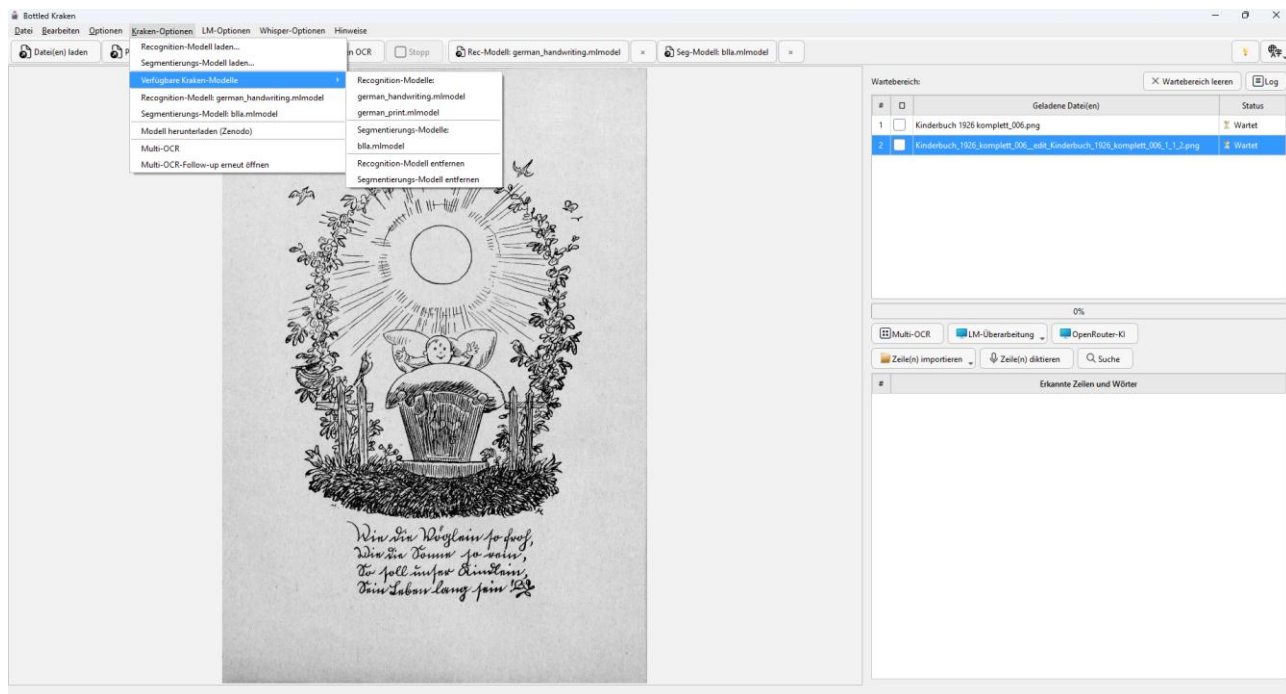
1. Start Bottled Kraken and load an image file or a PDF file



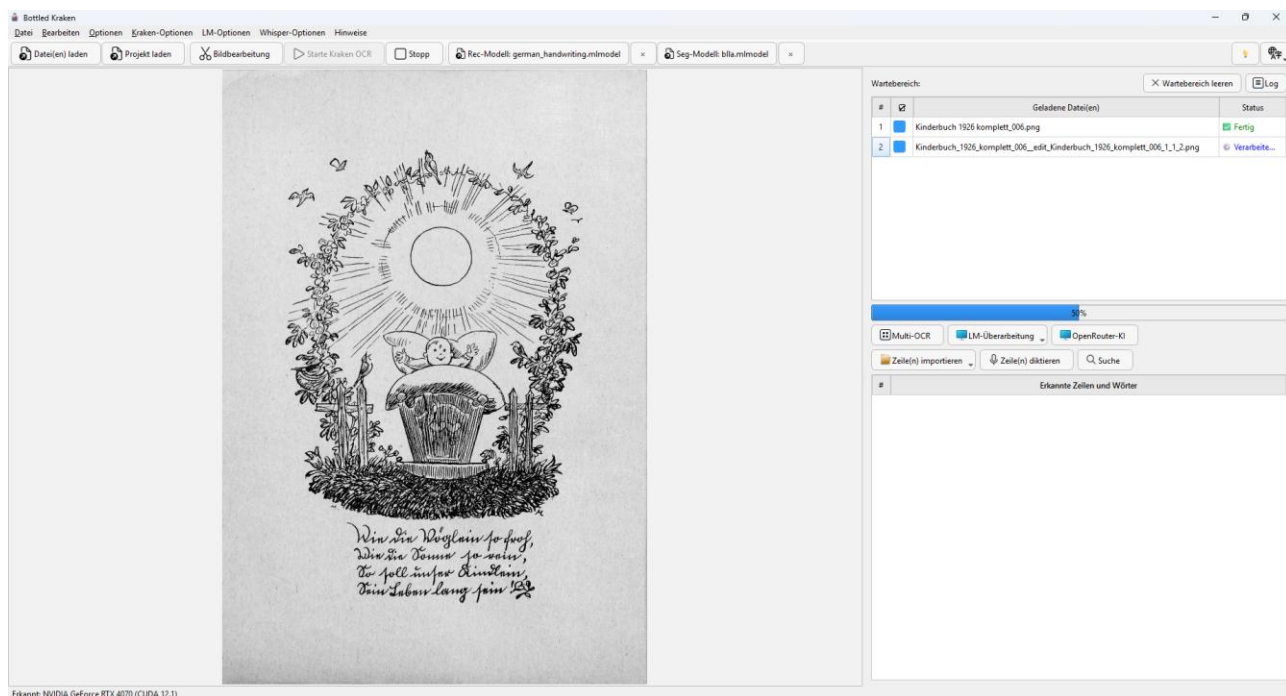
2. Start image editing and adjust the image (tip: grayscale is suitable for Kraken OCR – colored images are suitable for LM revision (AI))



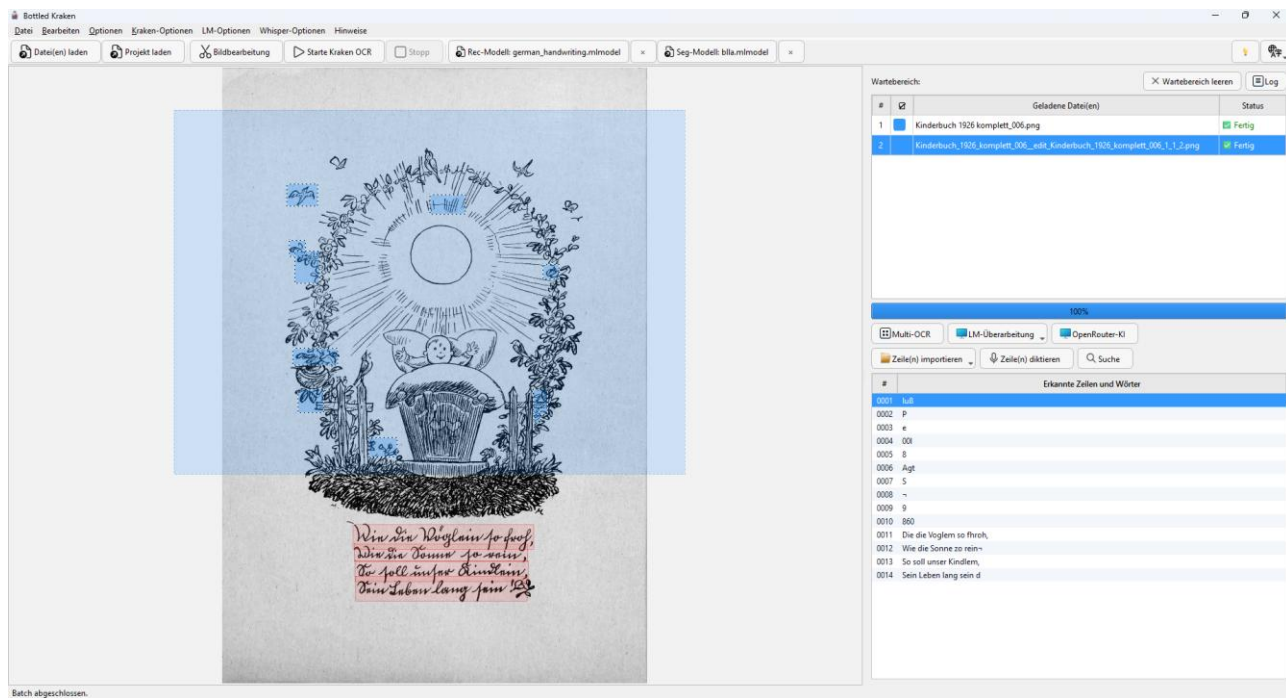
3. Loading Kraken Models



4. Start Kraken OCR (optional: select the files in the waiting area for which OCR should be performed)

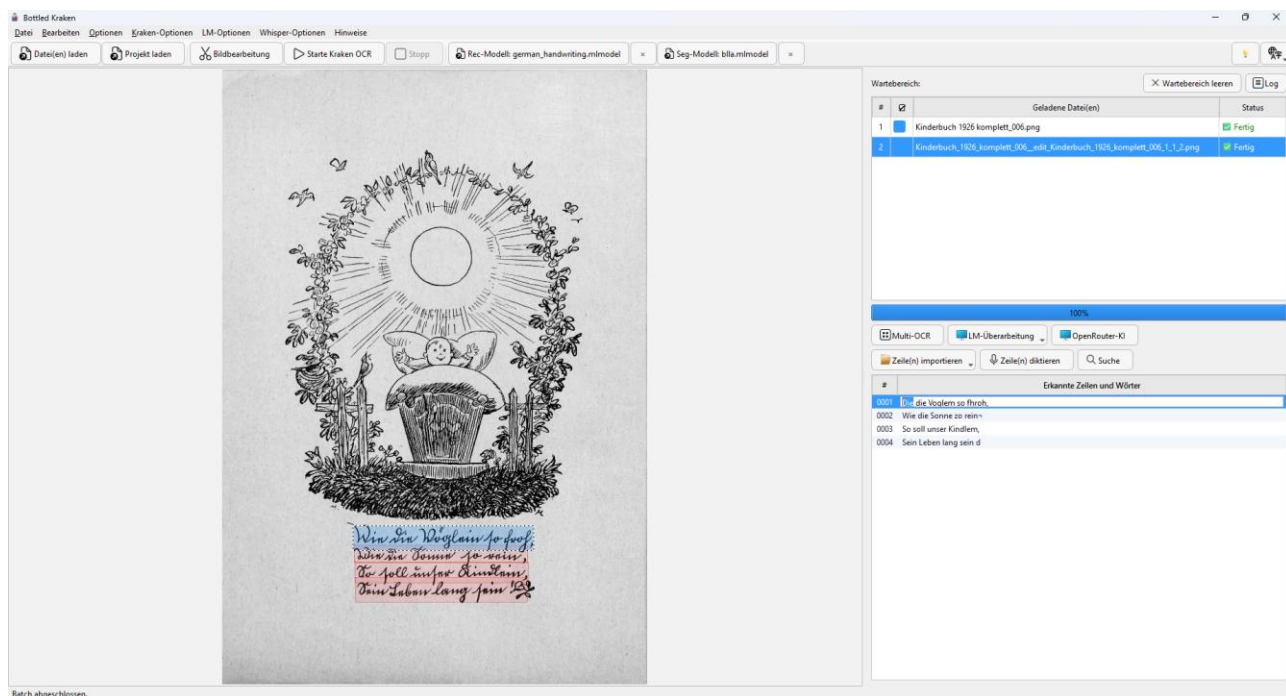


5. Adjusting/removing overlay boxes

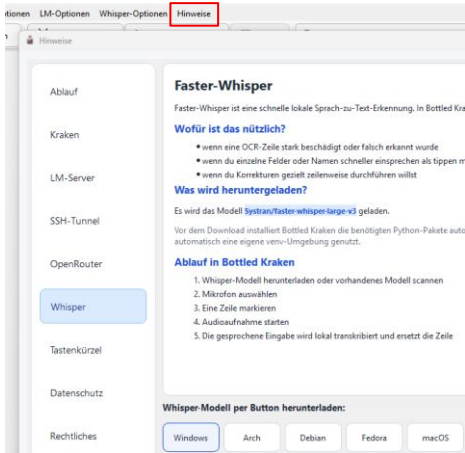


6. Adapt/change/revise content

a) Manually (in "Recognized Lines and Words")



b) with Whisper (downloadable in "Notes" > "Whisper" > "Operating System (e.g. Windows)")



Hinweise

Faster-Whisper

Faster-Whisper ist eine schnelle lokale Sprach-zu-Text-Erkennung. In Bottled Kraken

Wofür ist das nützlich?

- wenn eine OCR-Zeile stark beschädigt oder falsch erkannt wurde
- wenn du einzelne Felder oder Namen schneller einsprechen als tippen möchtest
- wenn du Korrekturen gezielt zeilenweise durchführen willst

Was wird heruntergeladen?

Es wird das Modell `systrain/faster-whisper-large-v3` geladen.

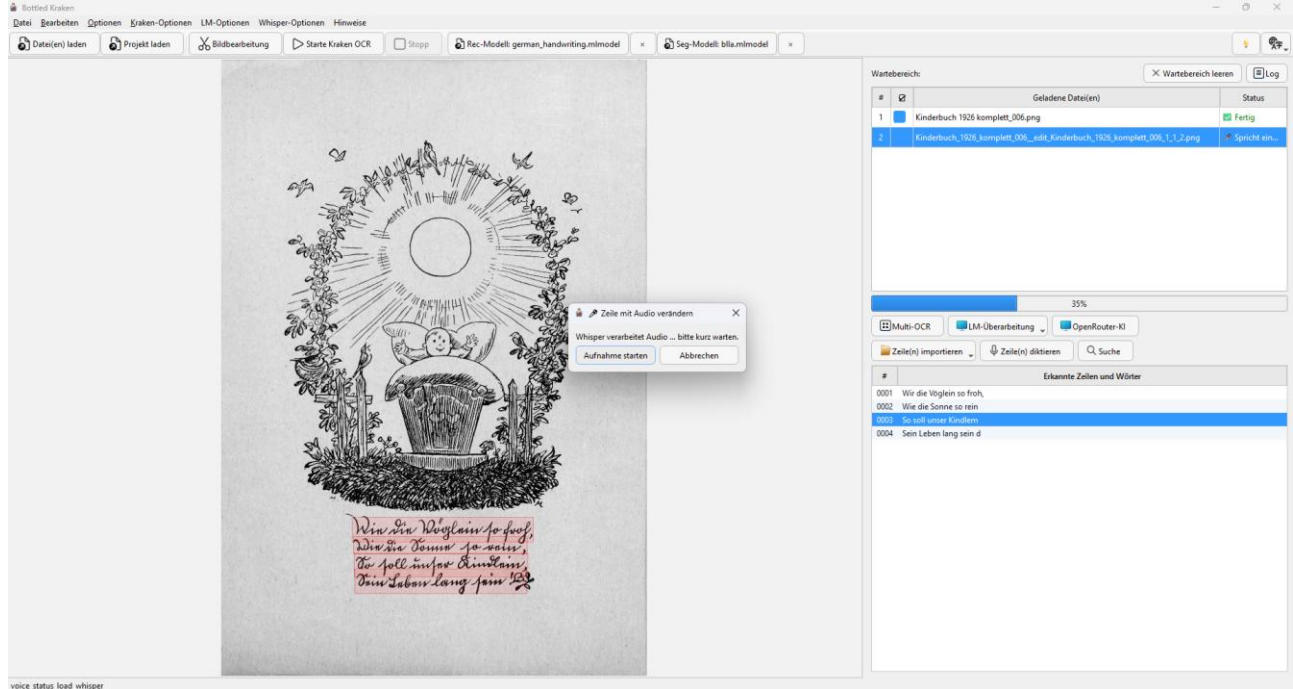
Vor dem Download installiert Bottled Kraken die benötigten Python-Pakete automatisch in eine eigene venv-Umgebung genutzt.

Ablauf in Bottled Kraken

1. Whisper-Modell herunterladen oder vorhandenes Modell scannen
2. Mikrofon auswählen
3. Eine Zeile markieren
4. Audioaufnahme starten
5. Die gesprochene Eingabe wird lokal transkribiert und ersetzt die Zeile

Whisper Modell per Button herunterladen:

Windows Arch Debian Fedora macOS



Wartebereich:

Wartebereich leeren Log

#	✓	Geladene Datei(en)	Status
1	☒	Kinderbuch_1926_komplett_006.png	Fertig
2	☒	Kinderbuch_1926_komplett_006_sdt_Kinderbuch_1926_komplett_006_v_1_2.png	• Spricht ein...

35%

Multi-OCR LM-Überarbeitung OpenRouter-KI

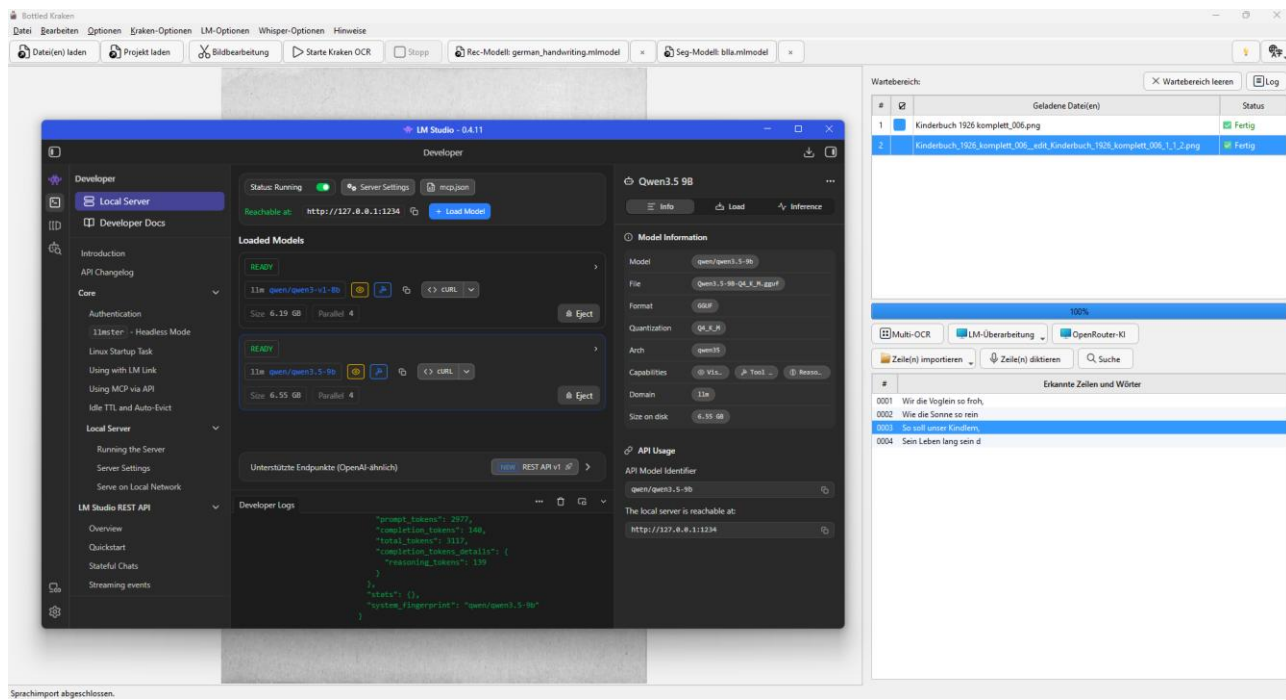
Zeile(n) importieren Zeile(n) diktieren Suche

Erkannte Zeilen und Wörter

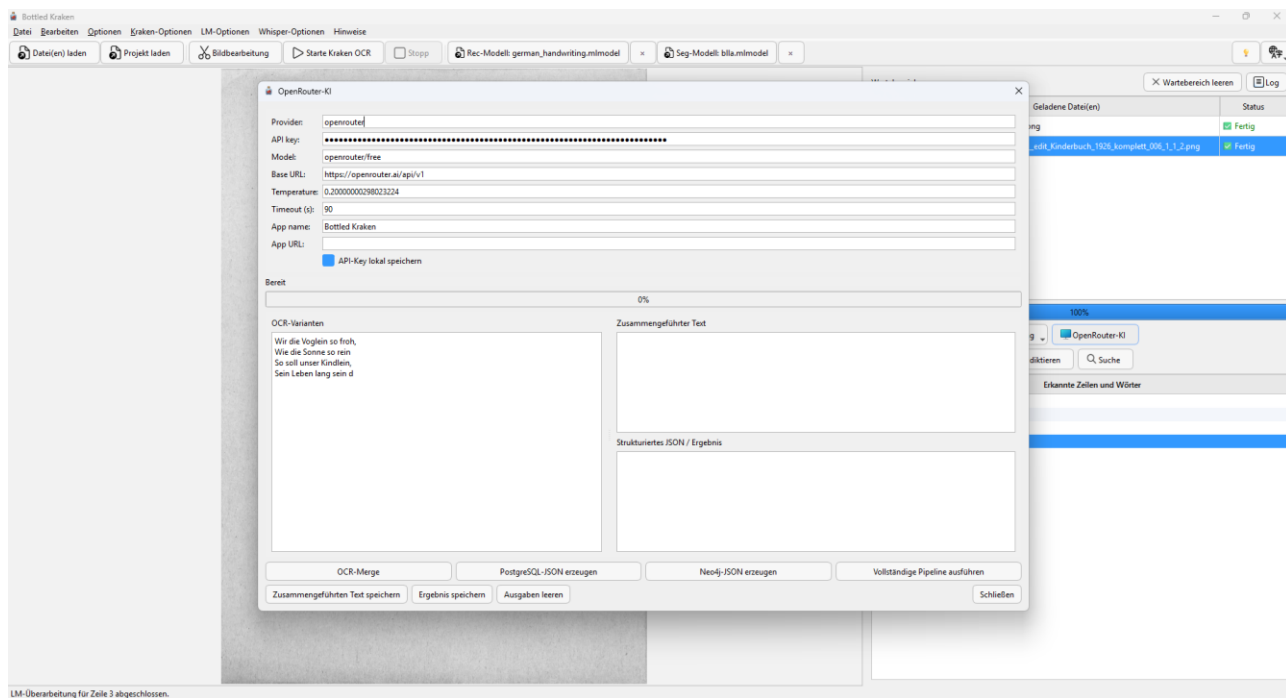
0001	Wie die Vögel so froh,
0002	Wie die Sonne so rein
0003	Du sollst unser Kind sein
0004	Dein Leben lang sein d

voice_status_load_whisper

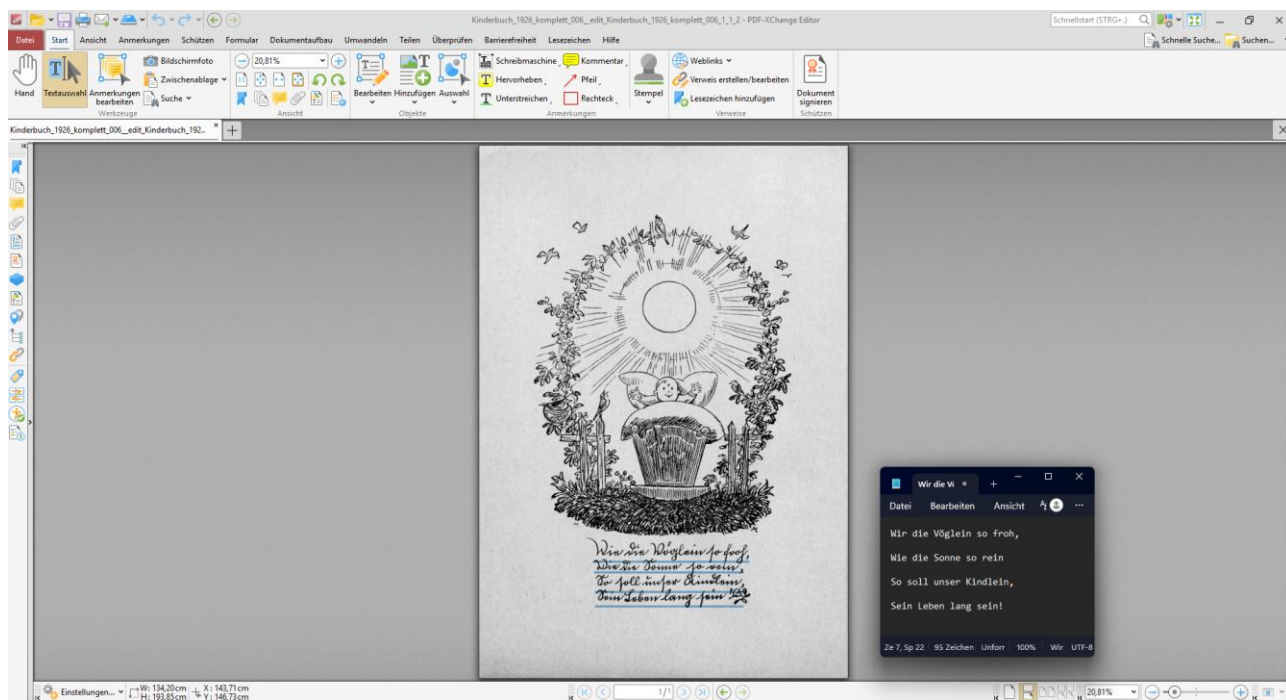
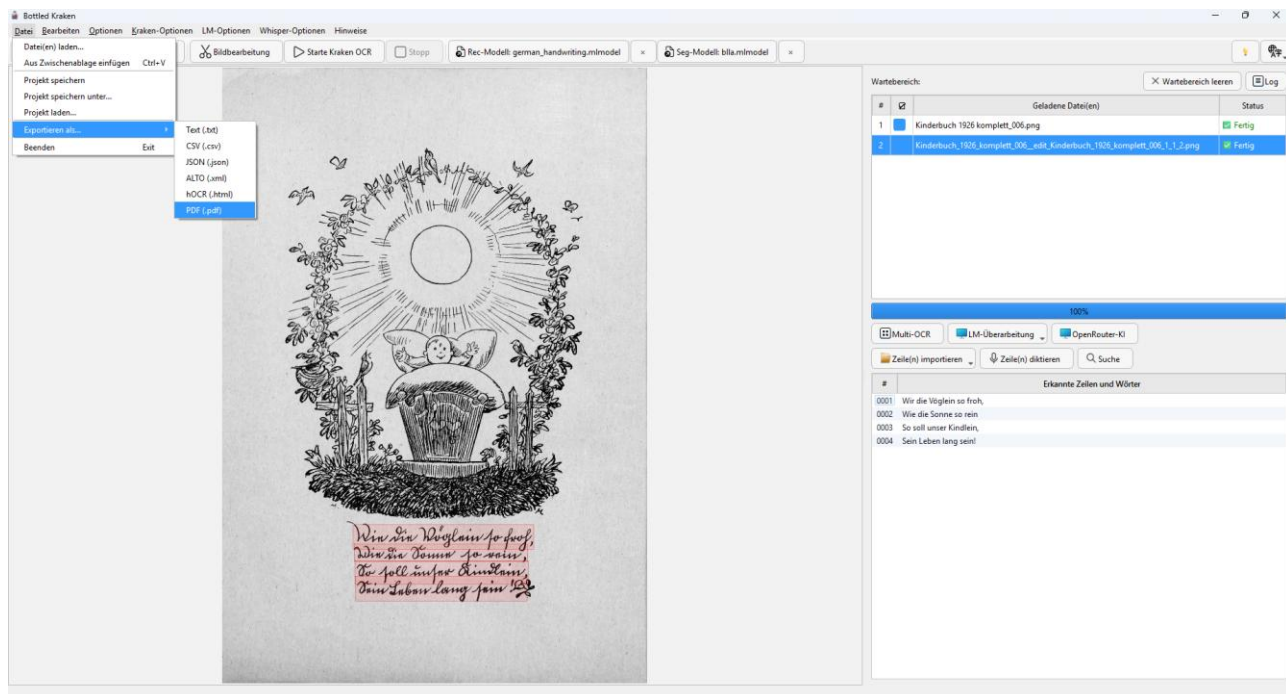
c) with a local AI model



d) with an external AI model (via OpenRouter)



7. Export



27. Glossary

Term	Short definition
Recognition	Text recognition within already identified lines.
Segmentation	Recognition of layout, lines, baselines and reading order.
Overlay-Box	Manually controllable area for targeted line correction.
Multi-OCR	Multiple OCR runs with different recognition models.
OCR-Merge	Merge several OCR variants into one target text.
PostgreSQL-JSON	Relationally oriented JSON structure for later table/ETL processes.
Neo4j-JSON	Graph-oriented JSON structure for nodes and relationships.
LM-Server	Local or self-hosted language model server with API.
API	Application programming interface through which Bottled Kraken communicates with local or external services.
SSH-Tunnel	Securely forwarding a remote service to a local port so that it can be used like a local server.
Kraken	Open-source OCR/ATR system used in Bottled Kraken for segmentation and text recognition.
Whisper	Speech-to-text model developed by OpenAI for automatic transcription of audio data.
faster-whisper	Fast local Whisper implementation based on CTranslate2, which is used in Bottled Kraken for microphone transcription.
Baseline	Imaginary baseline of a line of text that characters align with and that is important for segmentation and recognition.
Token	Smallest processing unit of a language model; Texts are processed internally in tokens instead of whole words.
CLI	Command-line interface that controls programs and models by terminal command instead of a graphical interface.

28. Acknowledgments

I would like to thank Benedikt E. for his support in the development of Bottled Kraken. He helped with the implementations of OpenRouter interfaces, the PostgreSQL and Neo4J implementations, and the conquest of the source code monolith.

Furthermore, I would like to thank Thomas E. (University of Leipzig) for his friendly and open encounter with this project, his patience and tolerance, his work as an alpha tester and feedback provider.

27. List of Sources

1. Primary Source: Bottled Kraken 3.0 Source Code
2. Bottled Kraken Repository: <https://github.com/Testatost/Bottled-Kraken>
3. Kraken Repository: <https://github.com/mittagessen/kraken>
4. Kraken Dokumentation: <https://kraken.re/7.0/index.html>
5. OpenRouter Quickstart: <https://openrouter.ai/docs/quickstart>
6. OpenRouter API Reference: <https://openrouter.ai/docs/api/reference/overview>
7. OpenRouter FAQ: <https://openrouter.ai/docs/faq>
8. OpenRouter Privacy Policy: <https://openrouter.ai/privacy>
9. OpenRouter Terms of Service: <https://openrouter.ai/terms>
10. OpenRouter Data Collection: <https://openrouter.ai/docs/guides/privacy/data-collection>
11. OpenRouter Provider Logging: <https://openrouter.ai/docs/guides/privacy/provider-logging>
12. LM Studio App Documentation: <https://lmstudio.ai/docs/app>
13. LM Studio Local API Server: <https://lmstudio.ai/docs/developer/core/server>
14. LM Studio OpenAI Compatibility: <https://lmstudio.ai/docs/developer/openai-compat>
15. LM Studio Privacy Policy: <https://lmstudio.ai/privacy>
16. LM Studio Terms: <https://lmstudio.ai/app-terms>
17. Ollama OpenAI Compatibility: <https://docs.ollama.com/api/openai-compatibility>
18. Ollama Cloud Models: <https://docs.ollama.com/cloud>
19. Jan Local API Server: <https://www.jan.ai/docs/desktop/api-server>
20. Jan Privacy: <https://www.jan.ai/docs/desktop/privacy>
21. GPT4All API Server: https://docs.gpt4all.io/gpt4all_api_server/home.html
22. LocalAI Overview: <https://localai.io/docs/overview/>
23. LocalAI Quickstart: https://localai.io/basics/getting_started/
24. faster-whisper Repository: <https://github.com/SYSTRAN/faster-whisper>
25. Qt for Python Documentation: <https://doc.qt.io/qtforpython-6/>
26. Qt for Python Lizenzübersicht: <https://doc.qt.io/qtforpython-6/licenses.html>
27. Martin Kässler: Local AI models on PCs with 8 GB VRAM: <https://www.martinkaessler.com/lokale-ki-modelle-auf-pcs-mit-8-gb-vram/>